Suricata IDS Lab - Command Cheat Sheet

Quick reference for the Suricata IDS Practical Lab | Authorized lab use only

> **Placeholders:** replace `<SURICATA_IP>`, `<CLIENT_IP>`, `<INTERFACE>` and `<LAB_SUBNET>` with your own lab values. Authorized lab use only.

Find your network details (both VMs)

Task	Command
List interfaces and IPs	`ip -br a`
Show routes	`ip route`
Interface used to reach the client	`ip route get <CLIENT_IP>`
Bring an interface up	`sudo ip link set <INTERFACE> up`
Test connectivity	`ping -c 3 <SURICATA_IP>`

Install and configure (Sensor VM)

Task	Command
Install	`sudo apt update && sudo apt install -y suricata jq netcat-openbsd`
Check version	`suricata -V`
Stop auto-started service	`sudo systemctl stop suricata`
Back up config	`sudo cp /etc/suricata/suricata.yaml /etc/suricata/suricata.yaml.bak`
Download standard rules	`sudo suricata-update`
Install lab rules	`sudo cp rules/local.rules /var/lib/suricata/rules/local.rules`
Edit config	`sudo nano /etc/suricata/suricata.yaml`
Validate config and rules	`sudo suricata -T -c /etc/suricata/suricata.yaml -v`

Run Suricata (Sensor VM)

Task	Command
Start in IDS mode	`sudo suricata -c /etc/suricata/suricata.yaml -i <INTERFACE>`
Start, checksums off (VM quirk)	`sudo suricata -c /etc/suricata/suricata.yaml -i <INTERFACE> -k none`
Reload rules live	`sudo kill -USR2 $(pidof suricata)`
Is Suricata running?	`pidof suricata`
Stop Suricata	`Ctrl+C` in its terminal, or `sudo pkill suricata`
Engine messages and errors	`sudo tail -n 30 /var/log/suricata/suricata.log`

Netcat traffic

Task	Command
Listener (Sensor)	`nc -lvnp 4444`
Auto-restarting listener	`./scripts/start_listener.sh 4444`

Task	Command
Check the listener port	`ss -ltnp | grep 4444`
Connection attempt (Client)	`nc -zv -w 2 <SURICATA_IP> 4444`
Send marker payload (Client)	`printf 'SURICATA-LAB-TEST hello\n' | nc -w 3 <SURICATA_IP> 4444`
All test traffic (Client)	`./scripts/generate_traffic.sh <SURICATA_IP> 4444`
Prove packets arrive	`sudo tcpdump -i <INTERFACE> -n port 4444 -c 5`

Read fast.log with grep

Task	Command
Watch alerts live	`sudo tail -f /var/log/suricata/fast.log`
Lab alerts only	`sudo grep "LAB " /var/log/suricata/fast.log`
One signature	`sudo grep "1000002" /var/log/suricata/fast.log`
Count alerts	`sudo grep -c "LAB Netcat" /var/log/suricata/fast.log`
One source IP, priority 1	`sudo grep "<CLIENT_IP>" /var/log/suricata/fast.log | grep "Priority: 1"`
Alerts per signature	`sudo grep -o '\[1:[0-9]*:[0-9]*\] [^[]*' /var/log/suricata/fast.log | sort | uniq -c | sort -rn`

Analyze eve.json with jq

Task	Command
Event types present	`sudo jq -r '.event_type' /var/log/suricata/eve.json | sort | uniq -c`
All alerts (pretty)	`sudo jq 'select(.event_type=="alert")' /var/log/suricata/eve.json`
Alerts by one SID	`sudo jq 'select(.alert.signature_id==1000002)' /var/log/suricata/eve.json`
Payload text	`sudo jq -r 'select(.alert.signature_id==1000002) | .payload_printable' /var/log/suricata/eve.json`
Source IP and signature	`sudo jq -r 'select(.event_type=="alert") | [.src_ip, .alert.signature] | @tsv' /var/log/suricata/eve.json`
Alerts from one IP	`sudo jq 'select(.event_type=="alert" and .src_ip=="<CLIENT_IP>")' /var/log/suricata/eve.json`
Full summary report	`sudo ./scripts/analyze_logs.sh`
Empty logs between runs	`sudo ./scripts/reset_logs.sh`

Rule syntax at a glance

```
alert tcp any any -> any 4444 (msg:"text"; content:"STRING"; nocase; sid:1000002; rev:1;)
```

Part	Meaning
`alert`	Action: log an alert (IDS mode).
`tcp` / `udp` / `icmp`	Protocol to inspect.
`any any -> any 4444`	Source IP and port, direction, destination IP and port.
`content:"..."` / `nocase`	Search the payload, ignoring case.
`flags:S`	Match the TCP SYN flag.
`sid` / `rev`	Unique rule ID (1000000 and above) and revision.

Log files (/var/log/suricata/)

File	Contains
`fast.log`	One-line text alerts.
`eve.json`	Rich JSON events: alerts, flows, stats.
`suricata.log`	Engine messages and errors.
`stats.log`	Performance counters.

Quick troubleshooting

Symptom	First checks
No route to host	Wrong IP or network, adapter down, firewall (`sudo ufw status`).
Connection refused	No listener: restart `nc -lvnp 4444`. Rule 1000001 still fires.
No alerts	Wrong interface (`ip route get <CLIENT_IP>`), rules not loaded (`suricata -T`), rule not in `rule-files`.
Rules failed to load	Check for a missing `;` or `)`, or a duplicate `sid`.
Alerts after edit missing	Copy the rule file again, then `sudo kill -USR2 $(pidof suricata)`.
Permission denied on logs	Use `sudo`.